

Week 11: Indexed expressions

Functional Programming in Lean

Learning goals

- ▶ Indexed inductive families
- ▶ Type interpretation with denote
- ▶ Typed evaluation

Core concepts

- ▶ Expr t: result type recorded by index
- ▶ denote .nat = Nat
- ▶ denote .bool = Bool

Lean example

```
inductive Expr : Ty → Type where
  | number (n : Nat) : Expr .nat
  | add (x y : Expr .nat) : Expr .nat
  | choose (c : Expr .bool)
    (x y : Expr t) : Expr t
```

Lean details

- ▶ add: two natural-number operands
- ▶ choose: branches at one index
- ▶ Rejected ill-typed constructor application

Live coding

- ▶ eval by pattern matching
- ▶ Typed conditional expression
- ▶ Rejected Boolean addition

Check

Result type of `eval .nat e`

Exercise

- ▶ Boolean expression
- ▶ Multiplication constructor